

Physiological biomarkers of stream fish from the Atlantic Forest within a Conservation Unit and its surroundings

Physiological biomarkers of fish as predictors of habitat quality in an Atlantic Forest stream Linking fish physiological biomarkers to habitat integrity in an Atlantic Forest Stream

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The Atlantic Forest is one of the most degraded biogeographical provinces of Brazil, and the creation of Conservative Unities is one of the main methods to reduce biodiversity loss in its extension. The project described here aims to compare the environmental conditions of streams, utilizing ecophysiological parameters that can indicate the presence of pollutants. For that, abiotic data and fish of a single genera (*Hemigrammus*) were collected inside the Guaribas Biological Reserve (Paraíba, Brazil) and outside it, subjected to different human interference. From the collected specimens, physiological markers were used as uniformity parameters for the distinct relationships between the fish from the study, being the quantification of total plasmatic proteins, moisture content of tissues, and the activity of MXR phenotype. Although the last one had inconclusive results, all of the parameters statistically demonstrated the existence of differences between the two groups and a more pronounced uniformity between the sites within the reserve. Combining this with the fact that fish from outside the reserve had a higher concentration of plasmatic proteins and also higher amounts of ammonia in the water of these sites, it was possible to argue that the conclusions regarding the effectiveness of the Conservative Unity were highly positive.

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1 Introduction

The Atlantic Forest is one of the world’s most important biodiversity hotspots (Myers et al., 2000). Despite occupying only a small fraction of its original distribution, it continues to support extraordinary levels of species richness and endemism (Marques et al., 2021). At the same time, it is one of the most threatened tropical ecosystems on Earth due to habitat loss, fragmentation, urban expansion, and agricultural development (Marques & Grelle, 2021). Conservation efforts in the Atlantic Forest have traditionally focused on terrestrial organisms such as plants, birds, and mammals. However, freshwater ecosystems are equally dependent on forest integrity and are often overlooked in conservation planning (Abilhoa et al., 2011; Gouveia et al., 2017).

Padial et al. (2021) mention a wide range of threats to aquatic systems that can affect the Atlantic Forest, ranging from dam construction and flow alterations to the emerging threat of microplastic contamination (Eerkes-Medrano & Thompson, 2018). Pollution, from pesticides, fertilizers, heavy metals, pharmaceuticals or microplastics (Evans, 1987; Harmon, 2018), come mostly from urban and agricultural sources, and untreated urban sewage is amongst the most common (Costa et al., 2020). In agricultural areas, fish farming, agricultural runoff, and the destruction of riparian forests, remain as important threats to aquatic ecosystem integrity (Padial et al., 2021).

Ecological responses to pollution, often become visible only after environmental impacts have already affected freshwater organisms (Macêdo et al., 2019). Furthermore, the toxicity responsible for behavioral change depends on the organism, time of exposure and particle type, which demonstrates the need for physiological studies, specially in natural environments (Amoatey & Baawain, 2019). For these reasons, physiological biomarkers can provide particularly valuable early-warning indication (Ondei et al., 2020), as physiological, morphological and behavioural changes os species are affected in the ecosystem.

Fish are among the most widely used organisms to evaluate the health of aquatic systems, since their their biochemical and physiological changes serve as biomarkers of environmental pollution (Cazenave et al., 2009; Ondei et al., 2020). One of the main physiological functions affected by pollutants is the osmoregulation, and its associated pH control. Both are predominantly done by gills, and are critical to gas exchange, ionic regulation and nitrogen excretion (Evans, 1987). The effects of this action on fish can be measured through a few blood markers, mainly lower levels of Chlorine and Sodium ions, obtained by freshwater fish through exchange with internal ions (Evans, 1987), and higher levels of plasmatic proteins (Sabae & Mohamed, 2015). The hydration level of gills and muscles is another physiological parameter associated with an animal's capacity to cope with environmental pollution and is therefore subject to measurable changes induced by toxic compounds present in the environment. This active control was suggested by Ayrapetyan (2012) as a universal biomarker for detecting pollution in the environment (David et al., 2018; Sabae & Mohamed, 2015).

Another relevant biomarker is the abnormal activity of the multixenobiotic resistance (MXR) phenotype (Bard, 2000), an innate gene of many aquatic species that is expressed in organs such as gills, kidneys, blood-brain barrier, liver and pancreas. It is activated by exposure to pollutants and results in the production of P-glycoproteins (Pgp) in the membrane, which act to remove endogenous or exogenous toxic compounds out of the cell, protecting DNA and inducing toxic compounds excretion (Smital & Kurelec, 1998). Although the MXR activity explains how some species are more resistant to the presence of pollutants, some substances can reverse its effectiveness, being called chemosensitizers (or simply MXR-inhibitors) (Smital et al., 2004). Those compounds can cancel the Pgp activity, alter the mechanism regulation, saturate or even reverse MXR activity, acting, in this case, by saturating the Pgp pump and allowing other xenobiotics to enter the cell and increase its toxicity, even if the chemosensitizers themselves are not toxic. Precisely because they are mostly organic, natural, primarily harmless and very present in the materials emitted anthropogenic in nature, its detection should also be a priority in environmental risk studies (Kurelec et al., 2000).

In the present study, the hypothesis of markers linked to physiological stress of fish providing evidence of the environmental stress was tested along a headwater stream that originates withing a protected area - Biological Reserve, or REBIO, according to BRASIL (2000) - and its course extends beyond its administrative boundaries (MMA, 2003). Given the importance of the Atlantic Forest to the biodiversity, we aim with this study, the identification of physiological stress biomarkers in fish from a headwater stream to evaluate the ecological health of sites inside and surrounding a protected area of Atlantic Forest, and enable decision makers to take

action on the influence of anthropogenic threats and the ecological consequence of pollution exposure of the buffer zone.

2 Material and Methods

Study area and sampling design. This study was conducted in the Guaribas Biological Reserve (REBIO Guaribas), a protected area of Atlantic Forest in northeastern Brazil. The reserve is located in the municipality of Mamanguape, state of Paraíba, in a heterogeneous landscape influenced by the adjacent Caatinga and Cerrado ecoregions. It was created in 1990 to preserve one of the last remnants of the Atlantic Forest in the region (MMA, 2003). Despite being one of the most restricted types of protected areas, there are a number human settlements in its buffer zone. The Management Plan for REBIO states several rural communities directly surrounding the reserve (Caiana, Pepina, Imbiribeira, Brejinho, Água Fria, João Pereira and Piabuçu), in addition to indigenous human populations (mostly the Potiguaras) (MMA, 2003). These populations are important because of their historic patterns of occupation and land use, but also because their agricultural activities, fishing, modification of the natural vegetation for livestock and overall use of the natural resources from the buffer zone directly affect the conservation of the REBIO. This has been aggravated in recent years by large scale use of water and land for cultivation of monocultures, like sugarcane (Heinrichs et al., 2017). The surrounding landscape is predominantly agricultural, including sugarcane and coconut plantations, livestock farming, and small orchards. These land uses can act as sources of diffuse pollution, especially through pesticide runoff and domestic effluents, creating a spatial gradient of environmental disturbance (MMA, 2003; Soler, 2004). Thus, the landscape is mainly rural and the land use causes direct impact to aquatic systems, by causing siltation of streams and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil (Arruda et al., 2013; Ministério do Meio Ambiente, 2025)

The climate is classified as As', characterized by a rainy season from February to July and a dry period from October to December (Peel et al., 2007). The average annual temperature varies from 24 to 26 °C, while the annual rainfall varies between 1,750 and 2,000 mm. The hydrographic network is composed of small streams fed by rain, notably the Barro Branco stream, a tributary of the Camaratuba River, where this study was carried out. These systems are influenced by riparian vegetation; however, parts of the buffer zone exhibit different levels of anthropogenic disturbance (Gouveia et al., 2017). Sampling was conducted along the Barro Branco stream, comprising six sampling sites, and at each site, three subsamples were collected between November 2023 and February 2024. The sampling sites were distributed longitudinally to capture environmental variability and gradients of human influence. Sites P1 to P3 were located within the reserve, where more preserved conditions were expected, while sites P4 to P6 were situated in the buffer zone or in adjacent areas with greater human influence (Figure 1). This design aimed to include different levels of habitat integrity, riparian condition, and water quality.

For the sake of comparison, we classified all sites into four categories defined *a priori* (Figure 2). Reference site or Type I site, which are the sites that occur within the conservation unit and are well preserved, with intact riparian forest and no crop fields nearby (at least 1 km distant), free flowing water (> 0.25 m/s), high dissolved oxygen concentration (> 6 mg/L) and temperature below 28 °C. Type II sites, are those that occur within the conservation unit, but are not well preserved, failing in one or more of the water quality criteria and/or having its riparian vegetation removed or managed. Type III sites are those outside the conservation unit but meet the water quality criteria for a type I site, and type IV sites are those that fall outside the conservation unit and are also disturbed, not meeting the water quality criteria for a type I site. Sites type III and IV were intentionally chosen not to meet the riparian forest and crop proximity criteria. Since they fall outside of the REBIO they are not expected to have intact riparian forest or to be at least 1 km distant from a crop field.

The ecophysiological evaluation of human impact on the forest, was mainly guided by the fish species inventory of the REBIO made by Gouveia et al. (2017). This study lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - *Oreochromis niloticus*, second most farmed fish species in the world and tolerant to osmotic variation, and *Poecilia reticulata*, largely used in mosquito control and resistant to stressful environments (Miranda, 2012). The predominance of small fish inhabiting small basins is thought to lead to low dispersion and, consequently, isolation of populations (Gouveia et al., 2017), associated with the fact that most species in the study area are phylogenetic close (with the predominance of Characidae) and are widely distributed, offers an appropriate design to access the entire local ichthyofauna using few species, namely the genera *Hemigrammus* and *Astyanax*.

Environmental data collection. Environmental characteristics of each sampling site were measured in four sets of variables: (a) site morphology, (b) water quality, (c) sediment composition, and (d) marginal habitat structure (Medeiros et al. (2008)). Site morphology was assessed by site width (cm) and depth (cm) and catchment scale variables (such as elevation and stream length) measured using handheld GPS and satellite imagery. Water quality was measured as physical and chemical variables using portable equipment for temperature (°C) and dissolved oxygen (mg/L). Transparency (cm) was measured using a Secchi disk and water velocity (m/s) was estimated using the float method Maitland (1990). Sediment composition and the habitat physical structure followed protocols adapted by Medeiros et al. (2008) from Pusey et al. (2004); Mugodo et al. (2006).

Fish collection and maintenance. Fish were collected between October and November 2023 along the Barro Branco Stream at six previously established sampling points, three located inside and three outside the Guaribas Biological Reserve, following a methodology adapted from Gouveia et al. (2017). Sampling was carried out during the day using a beach seine net (4 m long, 1.5 m high, and 5 mm mesh) and a hand net (50 cm wide and 5 mm mesh) operated with sweeping movements in shallow waters, according to Medeiros et al. (2010). Sampling effort was standardized across all collection events and sampling points.

To ensure the survival of specimens for ecophysiological analyses, the captured fish were not chemically fixed. Instead, they were immediately transferred to plastic bags containing water from the collection site and atmospheric oxygen, and subsequently transported to the Laboratory of Animal Ecophysiology (LEFA) at the State University of Paraíba, Campus V, João Pessoa, Brazil. Specimen sorting, taxonomic identification, and subsequent experimental procedures were performed at the Ecology Laboratory of the same institution, using specialized taxonomic literature and identification keys. The individuals were then separated according to the experimental groups established for the study.

For analyses related to osmoregulatory function, blood samples were obtained from 78 individuals, including 38 fish collected within the reserve and 40 in external areas. The fish were anesthetized with eugenol, and blood aliquots were collected by cardiac puncture using pipettes with heparinized tips. The samples were kept on ice until laboratory processing and subsequently centrifuged for plasma separation. Due to the reduced volume of plasma obtained as a consequence of the small body size of the specimens, chloride ion analysis could not be performed. Branchial and muscle tissues from the same individuals were collected to determine tissue moisture content. Plasma and tissue samples were stored at -20°C until laboratory analysis was performed.

The remaining 111 individuals, comprising 54 specimens from locations within the reserve and 57 from external areas, were used for evaluation of the multixenobiotic resistance (MXR) phenotype, following the methodologies described by David et al. (2018); Macêdo et al. (2019); Santos et al. (2017).

Plasmatic protein dosage. The plasma samples were defrosted and an aliquot of 2 μL of each was diluted with 8 μL of distilled water (proportion) to perform the total protein dosage by the Bradford (1976) method, in which the dye was also diluted in proportion. The intensity of each sample coloration was measured with a microplate reader (SpectraMax i3) at 595 nm wave length, and compared to a standard curve (BSA protein). The results were transferred to a spreadsheet, where the average value in milligrams of protein dosage was calculated for each sampling site, and a Kruskal-Wallis test was performed to determine the existence of differences between the averages of the points inside and outside the Reserva Biológica Guaribas, calculated in the R Studio program (RStudio Team, n.d.). Next, an analysis of variance (ANOVA) was performed to determine where there were differences between each sampling site.

Moisture content. The tissues samples (gill and muscle) were individually stored in 2 mL eppendorf tubes (previously weighted), initially weighted with the tissue moist, and, after 24 hours inside a 60°C oven, weighted with the dry tissue. The results were transferred to a spreadsheet, where it was calculated the percentual content of tissue water content and the average valor for each sampling site. A T-test (gill) or Kruskal-Wallis test (muscle) was performed to determine if there were significant differences between the average values inside and outside the Conservative Unity, followed by an ANOVA to determine the difference between all six sampling sites.

MXR phenotype activity. The multixenobiotic resistance (MXR) gene activity was analysed through the rhodamine B accumulation assay adapted from Smital & Kurelec (1998) (ref_fig). Rhodamine B is a substrate of P-glycoprotein, the molecular basis of the MXR phenotype. Thus, after exposing the fish to this substrate, the analysis of the amount of rhodamine accumulated in the animal's tissues (mainly the gills) reflects the activity of the MXR phenotype: the greater the accumulation, the lower the activity, and the lower the accumulation, the greater the activity. Therefore, specimens of fish from each sampling site (three inside the Conservative Unity and three outside, total of six sampling sites) were transferred to plastic aquariums containing dechlorinated water and rhodamine B at 2,5 μM concentration, staying in this condition for one hour. The aquarium remained under constant aeration and protected from direct light. After exposition, the fish were anesthetized with eugenol and sacrificed for gill sampling. The tissue samples were allocated in Eppendorfs tubes and weighted in an analytical scale. The weight of moist tissue was obtained by subtracting the weight of the previously weighted empty eppendorf tubes. The tissues were then homogenized with 500 μL of distilled water and the supernatant was transferred in triplicate of 100 μL to a 96-well microplate. The fluorescence intensity of the supernatant (corresponding to the intracellular rhodamine B fluorescence in the tissue) was measured in the microplate reader (SpectraMax i3) at 544 nm excitation and 590 nm emission. The fluorescence value was then normalized by the moist tissue weight in milligrammes, and a Kruskal-Wallis test was calculated in R Studio (RStudio Team, n.d.) to determine if there were differences between the averages inside and outside the Conservative Unity, followed by an ANOVA to visualize the differences between each sampling site.

Statistical analysis. Environmental variables were checked for multivariate collinearity and square root transformed before analyses (Sokal & Rohlf (1995)).

These variables were subsequently subjected to Principal Component Analysis (PCA) to evaluate multivariate correlations among sites. Prior to analysis, site morphology and water quality variables were square root transformed, whereas sediment composition and the marginal habitat structure (which were measured as percentages) were arcsine square-root transformed after relativization by column total McCune & Grace (2002). PCA was performed using the **FactoMineR** package in R Lê et al. (2008). All variables were centered and scaled to unit of variance.

3 Results

Environmental variables. Water flow was generally slow or absent across sampled streams (0.00-0.35 m/s). Stream sites tended to be narrow, with widths ranging from 0.66 to 16.0 m. Average depths varied from 9.0 to 51.3 cm across all sites. Dissolved oxygen (DO) ranged from 4.9 to 8.5 mg/L and temperatures from 24.9 to 28.5 °C. Waters were slightly acidic to neutral, with pH ranging from 4.5 to 7.7. Mud and sand were the main substrates across all sampled sites (with average covers of 72.4 and 27.6%, respectively), while coarse substrates

were absent. The habitat elements that gave greater overall contributions were overhanging terrestrial vegetation (53.8% on average), root masses (13.1%), littoral grass (11.1%) and leaf litter (11.1%), but these contributions varied widely across sites.

Principal Component Analysis described the overall structure of the study sites and the most important features in separating them in terms of their physical and chemical variables, site morphometry, sediment composition, and marginal habitat structure Figure 3. PCA explained 52.4% of the variance in the environmental variables, with the first axis (30.9%) showing a clear spatial gradient. This primary axis separated sites P08 and P10—characterized by greater stream width, depth, and sandy substrate—from sites P05, P06, and P07, which were associated with muddy substrates, higher DO, and dense overhanging vegetation. The second axis (21.5%) was primarily driven by local habitat features, separating site P09, which was strongly associated with root masses and water velocity, from sites characterized by attached algae and macrophytes. The overall multivariate conditions did not differ significantly between the interior and the surroundings of the reserve, although specific local variables such as water turbidity, leaf litter cover, and attached algae were important in explaining specific sites across the conservation unit's boundary.

```
1  ### limpar antes de começar
2  dev.off() #apaga os graficos, se houver algum
3  rm(list=ls(all=TRUE)) #limpa a memória
4  cat("\014") #limpa o console
5
6  ### verificar e definir diretório de trabalho
7  getwd()
8  setwd("D:/mxr23_Q")
9
10 ### carregar pacote e importar a matriz
11 library(openxlsx)
12 habitat <- read.xlsx("D:/mxr23_Q/data/rebio23-habitat.xlsx",
13                     rowNames = T,
14                     colNames = T,
15                     sheet = "ambiente",
16                     rows = 2:20)
17
18 ### substituir as na por 0
19 habitat[is.na(habitat)] <- 0
20 habitat
21
22 ### remove as colunas zeradas
23 sum <- colSums(habitat)
24 sum
25 zero_sum <- names(which(colSums(habitat) == 0))
```

```

26 zero_sum #nomes das colunas zeradas
27 m_part_cols <- habitat[(colSums(habitat) != 0)] #em != a exclamação inverte o sentido
28 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
29 zero_sum2 #nomes das colunas zeradas
30
31 ### recalcular soma das colunas depois de tirar as que estão zeradas
32 sum<-colSums(m_part_cols)
33 sum
34
35 ### importar matriz de grupos
36 t_grps <- read.xlsx("D:/mxr23_Q/mxr23_Q/data/rebio23-habitat.xlsx",
37                     rowNames = T,
38                     colNames = T,
39                     sheet = "grupos",
40                     rows = 2:20)
41 t_grps
42
43 ### salvar matriz de habitat filtrada
44 write.table(m_part_cols, "m_hab.csv",
45             sep = ";", dec = ".", #"\t",
46             row.names = TRUE,
47             quote = TRUE,
48             append = FALSE)
49
50 ### salvar matriz de grupos
51 write.table(t_grps, "t_grps.csv",
52             sep = ";", dec = ".", #"\t",
53             row.names = TRUE,
54             quote = TRUE,
55             append = FALSE)
56
57 ### reimportar matriz de grupos
58 t_grps <- read.csv("t_grps.csv",
59                   sep = ";", dec = ".",
60                   row.names = 1,
61                   header = TRUE,
62                   na.strings = NA)
63
64 ### reimportar matriz de habitat
65 m_hab <- read.csv("m_hab.csv",
66                  sep = ";", dec = ".",
67                  row.names = 1,
68                  header = TRUE,

```

```

69         na.strings = NA)
70
71 ### correlograma
72 library(psych)
73 colnames(m_hab)
74 #png("fig-h.hab_pairs.png")
75 pairs.panels(m_hab[,15:24],
76             method = "pearson", # correlation method
77             scale = FALSE, lm = FALSE,
78             hist.col = "#00AFBB", pch = 19,
79             density = TRUE, # show density plots
80             ellipses = TRUE, # show correlation ellipses
81             alpha = 0.5)
82 dev.off()
83
84 ### calcular matriz de correlação
85 cor <- cor(m_hab)
86 cor
87
88 ### plotar correlograma
89 library(corrplot)
90 #png("fig-hab_corrplot.png")
91 corrplot(cor, method = "circle")
92 dev.off()
93
94 ### deletando colineares ou irrelevantes
95 ### salvar saída das variáveis colineares
96 sink(file = "colineares.txt", append = F, split = T)
97
98 ### definir manualmente colunas a remover
99 colnames(m_hab)
100 del_cols <- c(
101   "w.Temp_C",
102   "w.Cond_uS.cm",
103   "w.pH",
104   "m.Slope",
105   "m.Depth_max_cm",
106   "m.Depth_mar_cm",
107   "s.Mud",
108   "h.Subm_veg",
109   "h.Leaf_l",
110   "h.Algae_f",
111   "h.Debris_l",

```

```

112 "h.Debris_s"
113 )
114
115 ### remover variáveis colineares
116 m_hab_part <- m_hab[, !(colnames(m_hab) %in% del_cols)]
117
118 ### códigos para somar redundantes (não foi necessário)
119 #m_hab_part$s.gravel <- m_hab_part$s.smlgrav + m_hab_part$s.lrggrav + m_hab_part$s.cobbles
120 #m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
121 #colnames(m_hab_part)
122 #m_hab_part
123 #sink()
124
125 ### visualizar nomes das variáveis filtradas
126 colnames(m_hab_part)
127
128 #####
129 # #
130 #Eu removi as variáveis de profundidade porque elas estavam muito correlacionadas #
131 #entre si (acima de 0,9), então basicamente estavam dizendo a mesma coisa. #
132 #Pra evitar redundância, deixei só a profundidade média. #
133 # #
134 #No caso do substrato, eu preferi manter sand em vez de mud #
135 #porque sand junto com turb foi o que melhor explicou a separação #
136 #no ponto 10 que era o mais degradado (Tipo IV). #
137 # #
138 #Já os pontos de referência (Tipo I) tinham mais variáveis influenciando, #
139 #não era uma coisa tão concentrada. #
140 # #
141 #As outras variáveis acabaram tendo vetores bem curtos na PCA, #
142 #ou seja, contribuíam muito pouco pra explicar os padrões. #
143 #A maioria tinha contribuição abaixo de 0,5, então já não fazia muito #
144 #sentido manter. Inclusive, várias dessas já tinham sido removidas antes #
145 #no seu script por serem pouco relevantes ou redundantes. #
146 # #
147 #####
148
149 ### transformação da matriz para fazer a PCA
150 ### aplicar transformação raiz quadrada
151 m_hab_trns <- sqrt(m_hab_part)
152
153 ### Fazendo a PCA
154 #install.packages("factoextra")

```

```

155 library("FactoMineR")
156 library("factoextra")
157
158 ### executar PCA
159 pca <- PCA(m_hab_trns, scale.unit = TRUE, ncp = 5, graph = TRUE)
160 print(pca)
161
162 ### extrair autovalores
163 eig.val <- get_eigenvalue(pca)
164 eig.val
165
166 ### salvar correlação das variáveis com os eixos
167 sink(file = "pca_cor_matrix.txt", append = FALSE)
168 pca$var$cor
169 sink()
170
171 ### gráfico de variância explicada
172 fviz_eig(pca, addlabels = TRUE)
173
174 ### extrair variáveis da PCA
175 var <- get_pca_var(pca)
176 var
177
178 ### plot das variáveis
179 fviz_pca_var(pca, col.var = "black")
180
181 ### visualizar cos2
182 library("corrplot")
183 corrplot(var$cos2, is.corr=FALSE)
184
185 ### gráfico cos2
186 fviz_cos2(pca, choice = "var", axes = 1:2)
187 ### plot com cores por cos2
188 fviz_pca_var(pca, col.var = "cos2",
189             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
190             repel = TRUE
191 )
192 ### transparência por cos2
193 fviz_pca_var(pca, alpha.var = "cos2")
194 ### contribuição das variáveis
195 corrplot(var$contrib, is.corr=FALSE)
196
197 ### variáveis mais importantes

```

```

198 fviz_contrib(pca, choice = "var", axes = 1, top = 10)
199 fviz_contrib(pca, choice = "var", axes = 2, top = 10)
200 fviz_contrib(pca, choice = "var", axes = 1:2, top = 10)
201 ### plot por contribuição
202 fviz_pca_var(pca, col.var = "contrib",
203               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07")
204             )
205 ### transparência por contribuição
206 fviz_pca_var(pca, alpha.var = "contrib")
207
208 ### clusterização das variáveis
209 set.seed(123)
210 res.km <- kmeans(var$coord, centers = 3, nstart = 25)
211 grp <- as.factor(res.km$cluster)
212
213 ### plot com clusters
214 fviz_pca_var(pca, col.var = grp,
215               palette = c("#0073C2FF", "#EFC000FF", "#868686FF"),
216               legend.title = "Cluster")
217
218 ### extrair os habitats
219 ind <- get_pca_ind(pca)
220 ind
221 ind$contrib
222
223 ### gráficos com os habitats
224 fviz_pca_ind(pca)
225
226 fviz_pca_ind(pca, col.ind = "cos2",
227               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
228               repel = TRUE
229             )
230
231 fviz_pca_ind(pca, pointsize = "cos2",
232               pointshape = 21, fill = "#E7B800",
233               repel = TRUE
234             )
235
236 fviz_pca_ind(pca, col.ind = "cos2", pointsize = "cos2",
237               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
238               repel = TRUE
239             )
240

```

```

241 ### qualidade de cada habitat
242 fviz_cos2(pca, choice = "ind")
243
244 ### contribuição de cada habitat
245 fviz_contrib(pca, choice = "ind", axes = 1:2)
246
247 ### variável de agrupamento
248 nrow(m_hab_part)
249 set.seed(123)
250 my.cont.var <- t_grps$Ref_site
251 my.cont.var
252
253 ### plot por grupo
254 fviz_pca_ind(pca,
255             col.ind = as.factor(t_grps$Ref_site),
256             palette = "jco",
257             addEllipses = TRUE)
258
259 ### plot com elipses
260 fviz_pca_ind(pca,
261             geom.ind = "point",
262             col.ind = as.factor(my.cont.var),
263             palette = "grey",
264             addEllipses = TRUE,
265             legend.title = "Groups"
266 )
267
268 ### número de grupos
269 length(unique(my.cont.var))
270
271 ### elipses de confiança
272 fviz_pca_ind(pca,
273             geom.ind = "point",
274             col.ind = as.factor(t_grps$Ref_site),
275             palette = "grey",
276             addEllipses = TRUE, ellipse.type = "confidence",
277             mean.point = TRUE,
278             legend.title = "Groups")
279
280 ### biplots
281 fviz_pca_biplot(pca, repel = TRUE,
282                col.var = "#2E9FDF",
283                col.ind = "#696969"

```

```

284 )
285
286 fviz_pca_biplot(pca,
287                 col.ind = as.factor(t_grps$Ref_site), palette = "jco",
288                 addEllipses = TRUE, ellipse.type = "euclid",
289                 label = "var",
290                 col.var = "black", repel = TRUE,
291                 legend.title = "Species")
292
293 ### GRAFICO FINAL
294 ### biplot final
295 fviz_pca_biplot(pca, repel = TRUE,
296                 col.var = "red",
297                 col.ind = "black")
298 )
299
300 library(factoextra)
301 library(ggplot2)
302
303 ### garantir variável como fator
304 t_grps$Ref_site <- as.factor(t_grps$Ref_site)
305
306 ### biplot com pontos e texto
307 fviz_pca_biplot(pca, repel = TRUE,
308                 col.var = "red",
309                 col.ind = "black",
310                 geom = c("point", "text"))
311
312 ### FILTRANDO VETORES PRINCIPAIS
313 ### selecionar variáveis mais importantes
314 var_coord <- pca$var$coord[, 1:2]
315 keep_vars <- apply(abs(var_coord) > 0.5, 1, any)
316 keep_vars
317 vars_to_keep <- rownames(var_coord)[keep_vars]
318 vars_to_keep
319
320 ### criar PCA filtrada
321 pca_filt <- pca
322 pca_filt$var$coord <- pca$var$coord[vars_to_keep, , drop = FALSE]
323 pca_filt$var$contrib <- pca$var$contrib[vars_to_keep, , drop = FALSE]
324 pca_filt$var$cos2 <- pca$var$cos2[vars_to_keep, , drop = FALSE]
325
326 ### biplot final filtrado

```



```

327 library(ggrepel)
328 #png("fig-hab_PCA_fviz.png", width = 11, height = 9, units = "in", res = 400)
329 fviz_pca_biplot(pca_filt, repel = TRUE, repel.var = TRUE,
330                col.var = "red",
331                col.ind = "black",
332                geom = c("text")) +
333   geom_point(aes(shape = t_grps$Ref_site), size = 3) +
334   scale_shape_manual(values = c(15, 16, 0, 1, 2, 17)) +
335   theme_minimal()
336 dev.off()

```

Fish community. A total of 422 individuals were collected. Fish were distributed among 6 species, 2 families, and a single order Characiformes. Among the families recorded, Characidae was the richest with 5 species, *Astyanax bimaculatus*, *Astyanax fasciatus*, *Hemigrammus marginatus*, *Hemigrammus rodwayi*, and *Hemigrammus unilineatus*, while Crenuchidae was represented by only 1 species, *Characidium bimaculatum*, with a single specimen collected. The most abundant species were *Hemigrammus unilineatus* (45.2% of the individuals collected). Other abundant species were *Hemigrammus rodwayi* (23.7%) and *Hemigrammus marginatus* (18.2%), establishing *Hemigrammus* as the dominant genus in the study area. Regarding their spatial distribution, only 1 species (*H. unilineatus*) was present at the sites located inside the Reserva Biológica Guaribas, whereas all 6 species were recorded in the surrounding areas. Out of the total collected fish, 118 specimens were submitted to the rhodamine B accumulation assay (MXR), 78 were used for total plasmatic protein dosage and moisture content quantification, and 226 individuals died prior to the physiological procedures Table 1.

Plasmatic protein dosage. The average plasmatic protein concentration, parameter that is positively correlated to environmental stress, was of 32,2 mg in the sampling sites within the Reserva Biológica Guaribas (n = 18 plasma samples) and 80,1 mg in the outside sites (n=26), revealing an almost 150% higher concentration in the sites with greater human influence. The total number of plasmatic protein dosages (n=44) was smaller than the total number of collected fish for this part of the analysis (n=78) because of the difficulty on collecting blood from smaller specimens, in addition to the need to collect a minimum amount of plasma required by the Bradford method. After checking the non-normality of variance of the total protein data, a non parametric Kruskal-Wallis test confirmed the existence of statistically significant difference (p-value<0,005) between the data collected from fish inside and outside the Conservative Unity (Figure 4).

```

1 library(readxl)
2 library(gplots)
3 library(agricolae)
4 library(sciplot)
5 library(PMCMRplus)
6 library(corrplot)
7 library(psych)

```

```

8 library(scales)
9 library(visreg)
10 citation()
11 dir()
12 setwd("D:/mxr23_Q")
13 tcc=read.csv("Dados_TCC.csv", h=T, stringsAsFactors = T, dec=",")
14 #TP
15 hist(tcc$TP, probability = T)
16 lines(x=density(x=tcc$TP, na.rm=T, adjust=2)) #parece um pouco
17 shapiro.test(tcc$TP) #0,001 = não normal
18 bartlett.test(tcc$TP~tcc$Local) #0,000001765 = não homo ###kruskal
19 kruskal.test(tcc$TP~tcc$Local) #0,00000002328 = diferem
20
21 par(mfrow=c(1,1),mar=c(3,5,1,1),lwd=1)
22 boxplot(tcc$TP~tcc$Local)
23 boxplot(tcc$TP~tcc$Local,axes=F, varwidth=T, xlab="", ylab="", border="black", col=c("darkgreen", "darkred"))
24 box(bty="o", lwd=2)
25 axis(side=2, las=1, font=2)
26 nomes=c("INSIDE", "OUTSIDE")
27 pos=c(1,2)
28 axis(side = 1, at = pos, labels = nomes, las=1, font=9)
29 mtext(text="Total Plasmatic Proteins (mg)", side=2, line=3, cex=1, font=7)
30 legend("topleft", legend=c("Kruskal-Wallis test demonstrated", "statistical difference", "(p<0.05)"))

```

The subsequent ANOVA test confirmed the uniformity of values from the sites inside the reserve (all grouped as “a”), differently from the outside sites (Figure 5).

```

1 anova(lm(formula=tcc$TP~tcc$Site)) #0,000000000000002783 = diferença
2 (tkPT=HSD.test(y=lm(formula=tcc$TP~tcc$Site), trt="tcc$Site", group=T))
3 boxplot(tcc$TP~tcc$Site) #a,a,a,c,b,bc #dentro e fora muito mais isolados
4
5 par(mfrow=c(1,1),mar=c(3,5,1,1),lwd=0.75)
6 boxplot(tcc$TP~tcc$Site,axes=F, varwidth=T, xlab="", ylab="", border="black", col=c("darkgreen", "darkred"))
7 box(bty="o", lwd=2)
8 axis(side=2, las=1, font=2)
9 nomesP=c("P5", "P6", "P7", "P8", "P9", "P10")
10 posP=c(1,2,3,4,5,6)
11 axis(side = 1, at = posP, labels = nomesP, las=1, font=9)
12 mtext(text="Total Plasmatic Protein (mg)", side=2, line=3, cex=1, font=7)
13 tapply(tcc$TP, tcc$Site, max, na.rm=T)
14 text(x=c(1,2,3,4,5,6), y=c(34,35.2,35.6,98,98,87.7)+7, labels=c("a", "a", "a", "c", "b", "bc"), cex=1.5)

```

The increase of total plasmatic protein concentration in the animals exposed to pollution is consistent with the (Sabae & Mohamed, 2015) study, which attributes this increase to five

possibilities: activation of metabolic systems as answer to pollutants exposure, degradation of cellular material in the liver, severe pathological conditions in the liver and kidneys, loss of water in the plasma, and induction of proteic synthesis in the liver.

Moisture content. The average values of moisture content, parameter related to the osmoregulatory function, were 84,3% for the gill and 80,3% for the muscle of fish inside the REBIO Guaribas (n=38 fish). Regarding the fish from sampling sites surrounding the reserve (n=40), the averages were 88,9% for the gills and 82,3% for the muscle, increasing, in order, only about 5,6% and 2,5% the values from inside, results that are not very expressive, like the ones described by Haredi et al. (2020). However, through T test (gills, which data had normal variation) and Kruskal-Wallis test (muscle, with non-normal variation data), it was detected statistical difference for both tissues between fish from the sampling sites inside and outside the Guaribas (Figure 6).

```

1  #TH_B = GMC
2  hist(tcc$GMC, probability = T)
3  lines(x=density(x=tcc$GMC, na.rm=T, adjust=2)) #quase
4  shapiro.test(tcc$GMC) #0,06 = normal
5  bartlett.test(tcc$GMC~tcc$Local) #0,7455 = homo ###teste t
6  t.test(tcc$GMC~tcc$Local) #0,002728 = diferem
7
8  par(mfrow=c(2,1),mar=c(2,5,1,1),lwd=1)
9  boxplot(tcc$GMC~tcc$Local)
10 boxplot(tcc$GMC~tcc$Local,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgray",
11 box(bty="o", lwd=2)
12 axis(side=2,las=1,font=2)
13 nomes=c("INSIDE","OUTSIDE")
14 pos=c(1,2)
15 axis(side = 1,at = pos, labels = nomes, las=1,font=9)
16 mtext(text="Gill Moisture", side=2,line=3.5,cex=0.9,font=7)
17 mtext(text="Content (%)", side=2,line=2.7,cex=0.9,font=7)
18 legend("top", legend=c("p-value=0,002728"), text.font = 2, bty="n", cex=0.55, text.col="gray")
19
20 #MMC
21 hist(tcc$MMC, probability = T)
22 lines(x=density(x=tcc$MMC, na.rm=T, adjust=2)) #parece
23 shapiro.test(tcc$MMC) #0,002 = não normal
24 bartlett.test(tcc$MMC~tcc$Local) #0,047 = não homo ###kruskal
25 kruskal.test(tcc$MMC~tcc$Local) #0,000004813 = diferem
26
27 par(mfrow=c(1,1),mar=c(1,5,5,1),lwd=1)
28 boxplot(tcc$MMC~tcc$Local)
29 boxplot(tcc$MMC~tcc$Local,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgray",

```

```

30 box(bty="o", lwd=2)
31 axis(side=2,las=1,font=2)
32 nomes=c("INSIDE","OUTSIDE")
33 pos=c(1,2)
34 axis(side = 1,at = pos, labels= nomes, las=1,font=9)
35 mtext(text="Muscle Moisture", side=2,line=3.5,cex=0.9,font=7)
36 mtext(text="Content (%)", side=2,line=2.7,cex=0.9,font=7)
37 legend("top", legend=c("p-value=0,000004813"), text.font = 2, bty="n", cex=0.55, text.col="g")

```

The ANOVA test demonstrated similarity between data from sites inside REBIO Guaribas (P5, P6 and P7) and P8, and a lesser similarity with P9, that was also close to P10 (Figure 7).

```

1 anova(lm(formula=tcc$GMC~tcc$Site)) #0,0000004971 = diferença
2 (tkTHB=HSD.test(y=lm(formula=tcc$GMC~tcc$Site),trt="tcc$Site",group=T))
3 boxplot(tcc$GMC~tcc$Site) #a,a,ab,a,bc,c ##P8 próximo aos de dentro, p9 e p10 mais isolados
4
5 par(mfrow=c(1,1),mar=c(2,5,1,1),lwd=0.75)
6 boxplot(tcc$GMC~tcc$Site,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgreen",
7 box(bty="o", lwd=2)
8 axis(side=2,las=1,font=2)
9 nomesP=c("P5","P6","P7","P8","P9","P10")
10 posP=c(1,2,3,4,5,6)
11 axis(side = 1,at = posP, labels = nomesP, las=1,font=9)
12 mtext(text="Gill Moisture", side=2,line=3.5,cex=0.9,font=7)
13 mtext(text="Content (%)", side=2,line=2.7,cex=0.9,font=7)
14 tapply(tcc$GMC,tcc$Site,max, na.rm=T)
15 text(x=c(1,2,3,4,5,6),y=c(91.3,94.3,98.1,89.1,97.1,97.9)+3,labels=c("a","a","ab","a","bc","c"))

```

The overall statistical analysis performed for moisture content reinforces this as a more discreet and inefficient parameter to make direct quantitative comparison for animals exposed to low salinity variation from the environment, but still accurate in determining the uniformity degree of more representative groups (i.e., fish from inside and outside the Conservative Unity).

MXR phenotype activity. The average values of normalized fluorescence measured in fish gills were 25.449 rfu/mg (relative fluorescence units/milligramme of tissue) for inside the Conservative Unity (n=54 fish) and 67.138,3 rfu/mg for outside (n=57), value nearly 2,64 times bigger. This difference between averages is opposite to the expected (higher fluorescence signals a lower MXR phenotype activity, meaning a lower exposition to pollutants), that can be explained by the presence of MXR phenotype inhibitors, which were highly correlated to more polluted waters by Kurelec et al. (2000). Among the possible pollutants presents, pesticides, fragrances, microbial degradation products and natural inhibitors from invasive species were described as compounds with high affinity to the Pgp or inhibiting the PKC (protein kinase C) modulator, resulting in high accumulation of rhodamine B (Smital et al., 2004) Despite the unexpected result, the variance difference between data from the MXR phenotype activity

inside and outside the REBIO Guaribas (confirmed by the Kruskal-Wallis test and illustrated in (Figure 8)

```

1  #MXR
2  hist(tcc$MXR, probability = T)
3  lines(x=density(x=tcc$MXR, na.rm=T, adjust=2)) #não parece
4  shapiro.test(tcc$MXR) #0,0000038 = não normal
5  bartlett.test(tcc$MXR~tcc$Local) #0,000264 = não homo ##kruskal
6  kruskal.test(tcc$MXR~tcc$Local) #0,0000000000005607 = diferem
7
8  par(mfrow=c(1,1),mar=c(3,6,1,1),lwd=1)
9  boxplot(tcc$MXR~tcc$Local)
10 boxplot(tcc$MXR~tcc$Local,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgreen", "darkred"))
11 box(bty="o", lwd=2)
12 axis(side=2,las=1,font=2)
13 nomes=c("INSIDE","OUTSIDE")
14 pos=c(1,2)
15 axis(side = 1,at = pos, labels = nomes, las=1,font=9)
16 mtext(text="Fluorescence/Mass of Tissue (rfu/mg)", side=2,line=4.5,cex=1,font=7)
17 legend("topleft", legend=c("Kruskal-Wallis test demonstrated", "statistical difference", "(p<0.05)"))

```

was also the most conclusive when analysed by ANOVA test, being the only physiological parameter that shows total uniformity for both relationship with the reserve (Figure 9).

```

1  anova(lm(formula=tcc$MXR~tcc$Site)) #0,00000000000005 = diferença
2  (tkMXR=HSD.test(y=lm(formula=tcc$MXR~tcc$Site),trt="tcc$Site",group=T))
3  boxplot(tcc$MXR~tcc$Site) #a,a,a,b,b,b #dentro e fora 100% isolados
4
5  dunnMXR=kwAllPairsDunnTest(formula=tcc$MXR~tcc$Site)
6  summary(dunnMXR) #ad,b,ab,c,c,cd
7
8  par(mfrow=c(1,1),mar=c(3,6,1,1),lwd=0.75)
9  boxplot(tcc$MXR~tcc$Site,axes=F, varwidth=T, xlab="",ylab="", border="black", col=c("darkgreen", "darkred"))
10 box(bty="o", lwd=2)
11 axis(side=2,las=1,font=2)
12 nomesP=c("P5","P6","P7","P8","P9","P10")
13 posP=c(1,2,3,4,5,6)
14 axis(side = 1,at = posP, labels = nomesP, las=1,font=9)
15 mtext(text="Fluorescence/Mass of Tissue (rfu/mg)", side=2,line=4.5,cex=1,font=7)
16 tapply(tcc$MXR,tcc$Site,max, na.rm=T)
17 text(x=c(1,2,3,4,5,6),y=c(68400,31371,44811,88000,150658,121050)+10000,labels=c("a","a","a","a","a","a"))

```

Abiotic data. The abiotic data from every sampling site water was measured and reunited in Tab. 1. The abnormal proportion of ammonia indicates the presence of effluents in the water

outside the Reserva Biológica Guaribas, and, as discussed by (Piedras et al., 2006), this elevated concentration generates metabolic ammonia retention, causing toxicity. The same paper also points out that the toxicity of non-ionized ammonia in the aquatic environment is greater for smaller organisms, but depends on the interaction with other environmental variables, such as temperature, pH and salinity, demanding specific studies with the target species to determine their tolerance to ammonia. The Pearson correlation analysis among biotic data showed only one significant correlation, between Total Protein and MXR phenotype activity (0,6 - moderate and positive), and among abiotic data only two significance, between Nitrate e Phosphor (0,69 - moderate and positive) and between Dissolved Oxygen and Ammonia (-0,82 - strong and negative). It was also detected significant correlation among both data, between Gill Moisture Content and Ammonia (0,55 - moderate and positive), between Gill Moisture Content and Dissolved Oxygen (-0,62 - moderate and negative), and between MXR activity and Nitrate (-0,62 - moderate and negative).

4 Discussion

The hypothesis of markers linked to physiological stress of fish providing evidence of the environmental stress levels occurring inside and outside of a Conservation Unity was tested in the Guaribas Environmental Reserve (REBIO Guaribas), one of the 22 CU's inside the Atlantic forest area of Paraíba state, located in the Zona da Mata, wettest and most economically relevant mesoregion from the state (Figure 1). Grouped, according to SNUC, as one of the integral protection areas (aimed to conserve biodiversity, being minority within the biogeographical province – 33,45% of the CU's), the definition of environmental reserves prohibits human interference inside their limits, except for management actions that aim its preservation and recovery. Despite being one of the most restricted types of Conservation Units, there are 22 populations that influence REBIO Guaribas due to historic factors of occupation and absence of an effective buffer zone around the three areas that make up the reserve. Those populations are mainly rural and cause direct impact, by using wood and plants from the reserve, and indirect, causing siltation of rivers and contaminating groundwater through the use of septic tanks and releasing agricultural chemicals into the soil, promoting possible comparable ecophysiological responses in the ichthyofauna inside and around the CU (Arruda et al., 2013; Ministério do Meio Ambiente, 2025).

The ecophysiological evaluation of human impact on the forest, which is close to areas of monoculture and natural vegetation modified for livestock, was mainly guided by the fish species inventory of REBIO Guaribas made by Gouveia et al. (2017), in a study that lists 18 species from five different orders: Characiformes (12 species), Perciformes (3 species), Synbranchiformes (1 species), Siluriformes (1 species) e Cyprinodontiformes (1 species). From those, two are exotic - *Oreochromis niloticus*, second most farmed fish species in the world and tolerant to osmotic variation, and *Poecilia reticulata*, largely used in mosquito control and resistant to stressful environments (Miranda, 2012) - and none are listed as threatened. The predominance of small fish inhabiting small basins explains the higher level of endemism in

streams, resulting from the allopatric speciation generated by the low dispersion made by these fish and the consequent isolation of populations (Gouveia et al., 2017). Having in mind that these stream fish are phylogenetic close, this paper offers representative data for the entire local ichthyofauna using species of *Hemigrammus* genus, of the Characiformes order, characterized by a black conspicuous spot delimited on the dorsal fin. Its type species, *Hemigrammus unilineatus*, has a compressed and slightly elongated body and exhibits wide geographical distribution both in Central and South America, with registers in other Brazilian states, such as Alagoas and Pernambuco.

All parameters were effective to point out statistical differences between data from inside the REBIO Guaribas and its surroundings, highlighting - alongside the higher levels of ammonia in the water and plasmatic proteins in the fish - that changes in the environment are impacting the fish from the streams studied. Beyond that, the statistical analysis of physiological parameters from fish collected inside the Conservation Unity showed uniform results among its three sampling sites, consistent with what is expected from natural environments with little human influence. The same did not occur among the three sites outside the CU (with the exception of the MXR phenotype activity), indicating the action of different levels of environmental stress. The differences between the average results from the physiological parameters measured inside and outside the Conservation Unity were illustrated in the Figure 10, highlighting the total protein as one that proved itself as an excellent biomarker to detect pollutants in the aquatic environment.

```

1 md=colMeans(tcc[grep("inside",x=tcc$Local),c(3:6)],na.rm = T)
2 mf=colMeans(tcc[grep(
3   "outside",x=tcc$Local),c(3:6)],na.rm = T)
4 (médias=matrix(c(md,mf),byrow=T,nrow = 2,ncol=4,dimnames=list(c("inside","outside"),c("GMC",
5   (prop=proportions(médias,2))
6   par(mfrow=c(1,1),mar=c(3,4,1,1),lwd=1)
7   barplot(height = prop, horiz = T, las=1, col = c("darkgreen", "darkred"), legend.text = T)

```

These preliminary obtained data were precise to indicate significant differences among the conditions in which the fish inhabited. But as much MXR phenotype activity data were as disparate as the ones of total protein and even more precise to point out the uniformity among sampling sites inside and outside the REBIO Guaribas, it showed values opposite to what was expected, with higher fluorescence in the tissue of fish outside the reserve, normally indicating lower MXR phenotype activity. This inconsistency can be explained by the possible presence of chemosensitizers in the water, since the fish did not went through a decontamination period, and thus, pollutants could still be bound to the active Pgp site and preventing rhodamine B to attach to it. The presence of such chemosensitizers can be quantified through the creation of a calibration curve, as described by (Kurelec et al., 2000), but, until done, the relevance of the MXR phenotype analysis must be restricted to the statistical context.

From the final results, an important positive evaluation can be made regarding the environmental protection that the Conservative Unity does within its extension, even when analysing one

with historical problems of delimitation and still not entirely free of anthropic impacts. Adding the fact that it is located within the Atlantic forest, classified as a Hotspot, the verification of the efficiency of a CU in this biogeographical province is even more relevant, reinforcing the need for governments to create more legally protected areas as a key strategy to reduce global ecological crisis. This study also made an important description on the use of physiological biomarkers in ecological analysis, raising ecophysiology potential as a basis to future assays, which can be incorporated into effectiveness index such as seen in (Masullo et al., 2020) paper and presenting quantifiable data that can complement studies with analysis of historical and geographical data, such as the one from (Assis et al., 2021).

Conclusions

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Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Adamastor Coutinho Pinto: Conceptualization, Data curation, Formal analysis, Investigation, Software, Visualization, Writing – review & editing, Writing – original draft.

Mayara Mirelly da Silva Monteiro: Data curation, Formal analysis, Software, Writing – review & editing.

Alice da Silva Barros: Data curation, Formal analysis, Software, Writing – review & editing.

Larissa Rafaela Caetano da Silva: Data curation, Formal analysis, Software, Writing – review & editing.

Enelise Marcelle Amado: Conceptualization, Data curation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Elvio Sergio Figueredo Medeiros: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing.

Ethical approval

Fish collection was authorized by the Instituto Chico Mendes de Conservação da Biodiversidade (license SISBIO-89415-2) and the project was approved by the Ethics Committee on Animal Use of the State University of Paraíba (protocol CEUA-048/2023).

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Figures and Tables

Figures

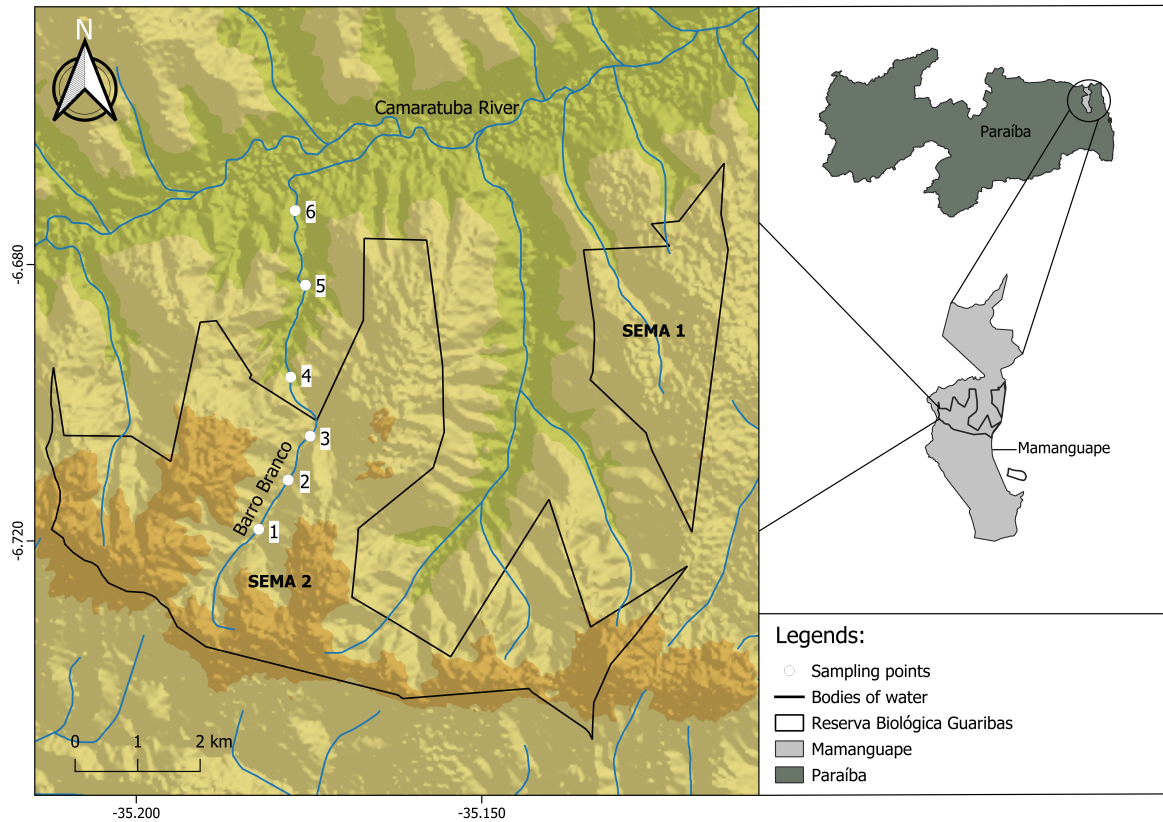


Figure 1: Location of sampling sites within (P1 - 06°43'06,58555'' S, 35°10'54,70041' W -, P2 - 06°42'38,71927'' S, 35°10'38,68013' W - and P3 - 06°43'06,58555'' S, 35°10'54,70041' W) and surrounding REBIO Guaribas (P4 - 06°41'46,29119'' S, 35°10'36,09933' W -, P5 - 06°40'56,47593'' S, 35°10'27,84963' W - and P6 - 06°40'18,97010'' S, 35°10'34,68148' W). In total, 92 fish were collected inside the Conservation Unity and 97 outside it.

Tables

Type I



Type II



Type III



Type IV



Figure 2: Images of the main four types of sites found in the Reserva Biológica Guaribas and its surrounding areas (Mamanguape, PB). Type I or local reference site, well preserved and within the conservation unit. Type II site, preserved and outside the conservation unit. Type III site, disturbed but withing the conservation unit. Type IV site, disturbed and outside the conservation unit.

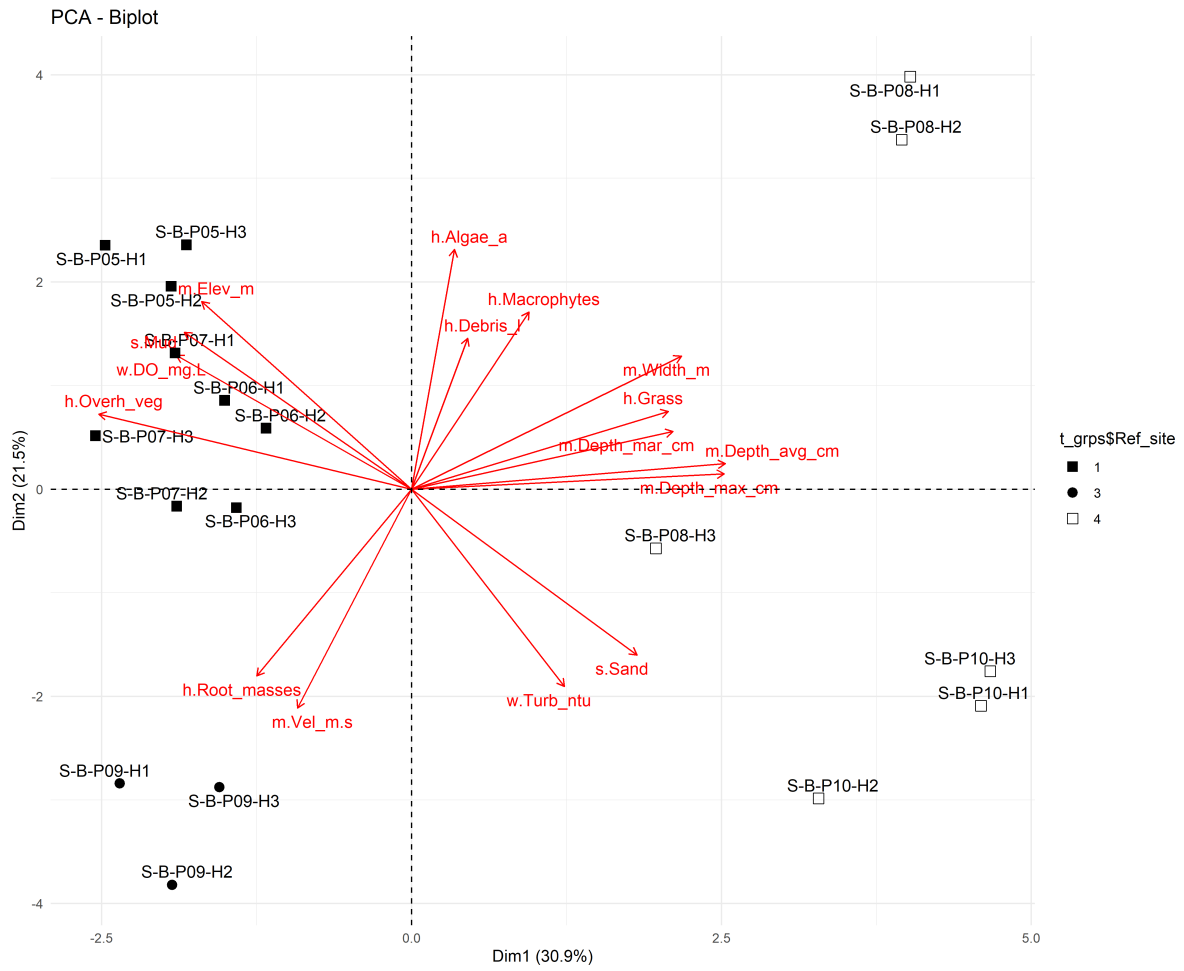


Figure 3: Principal Component Analysis of the environmental variables using the fviz function.

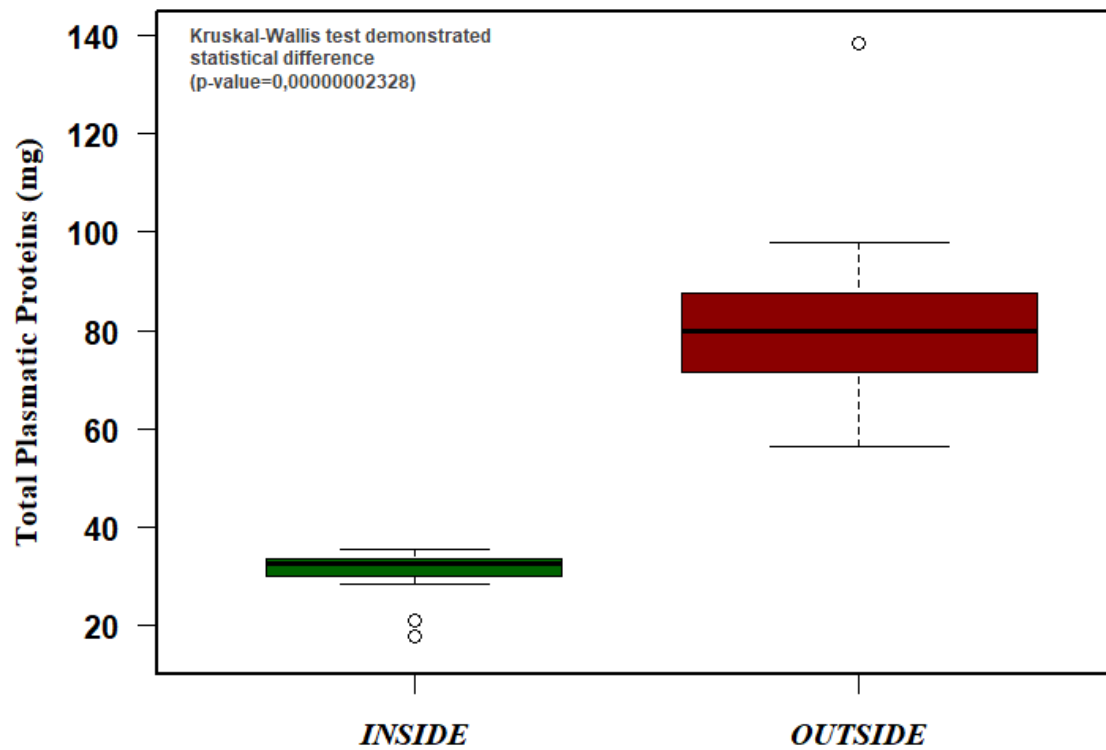


Figure 4: Total protein (mg) variance of data inside and outside REBIO Guaribas.

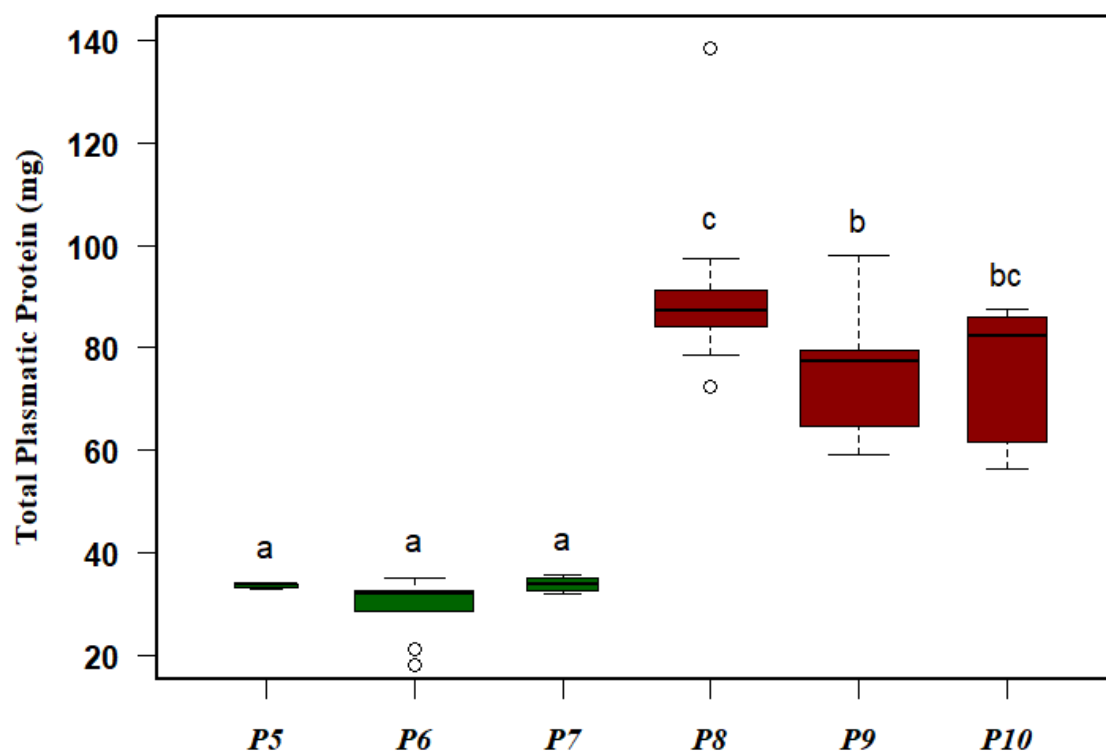


Figure 5: Boxplot with total protein data of collected fish from the sampling sites inside (green) and outside REBIO Guaribas (red), including its respective grouping by similarity above each box.

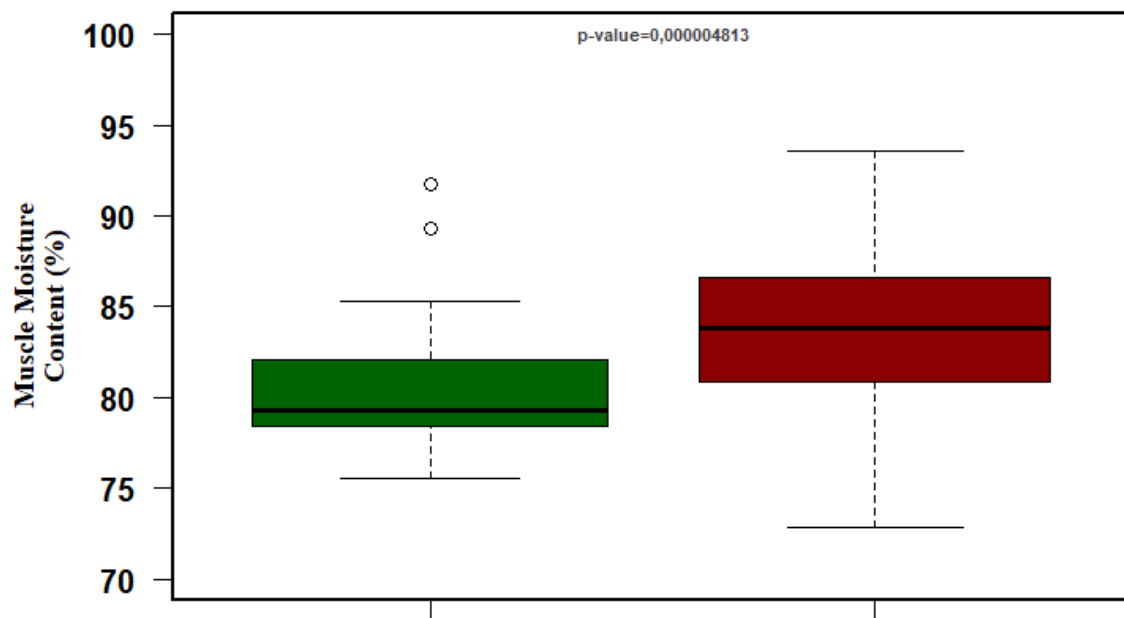


Figure 6: Moisture content (%) of tissues from sampling sites inside and outside the REBIO Guaribas, with the respective p-value between each locality.

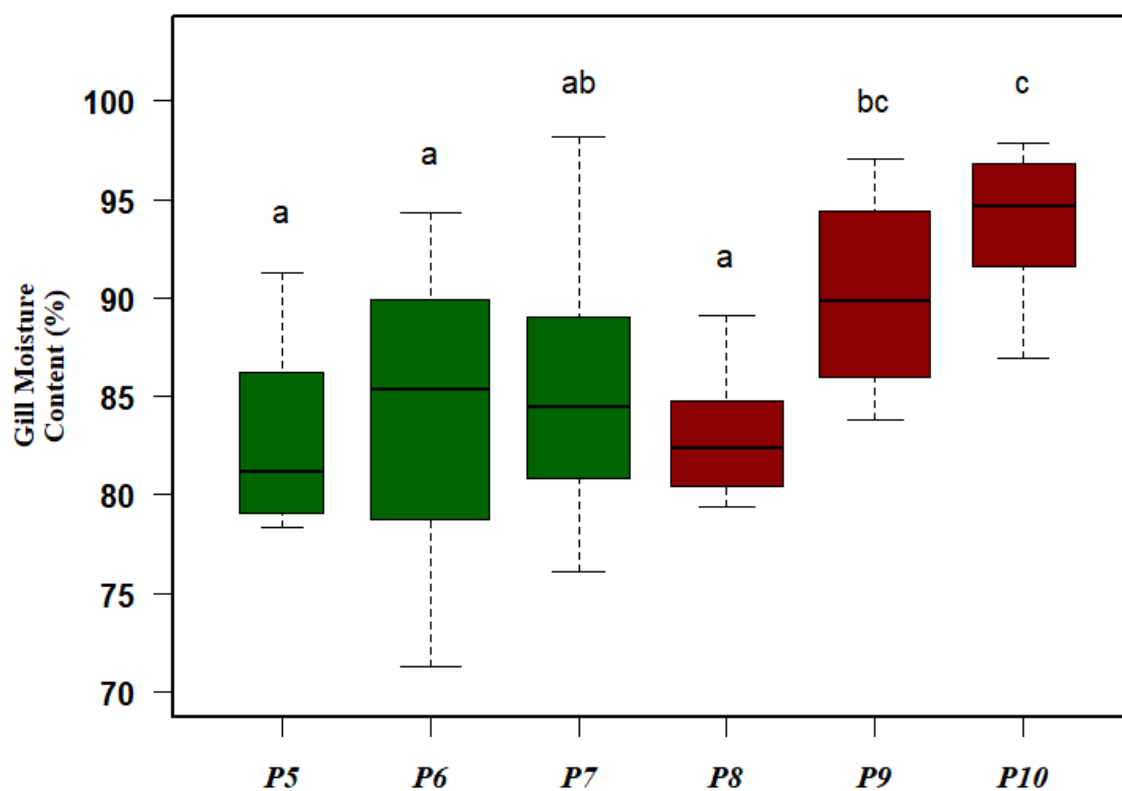


Figure 7: Boxplot with gill and muscle moisture content data of fish collected from sites inside (green) and outside REBIO Guaribas (red), with its respective grouping by similarity above each box.

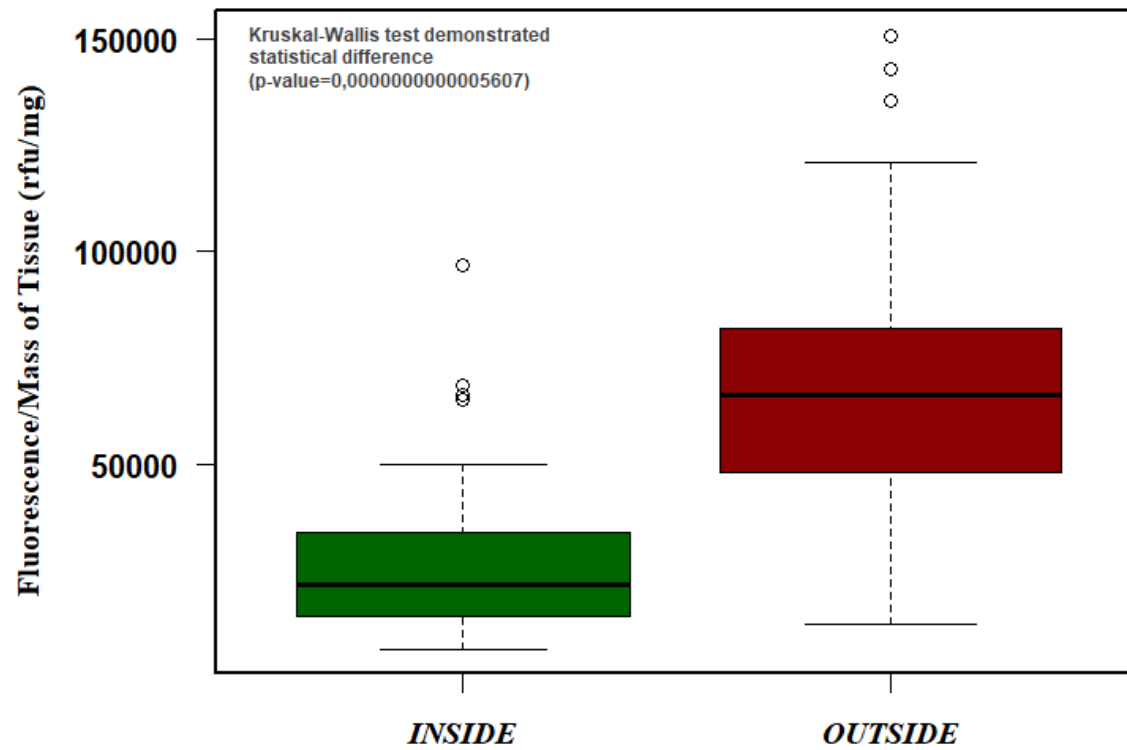


Figure 8: Average fluorescence data from sampling sites inside and outside the REBIO Guaribas, expressed by relative fluorescence units/milligrammes of tissue (gill).

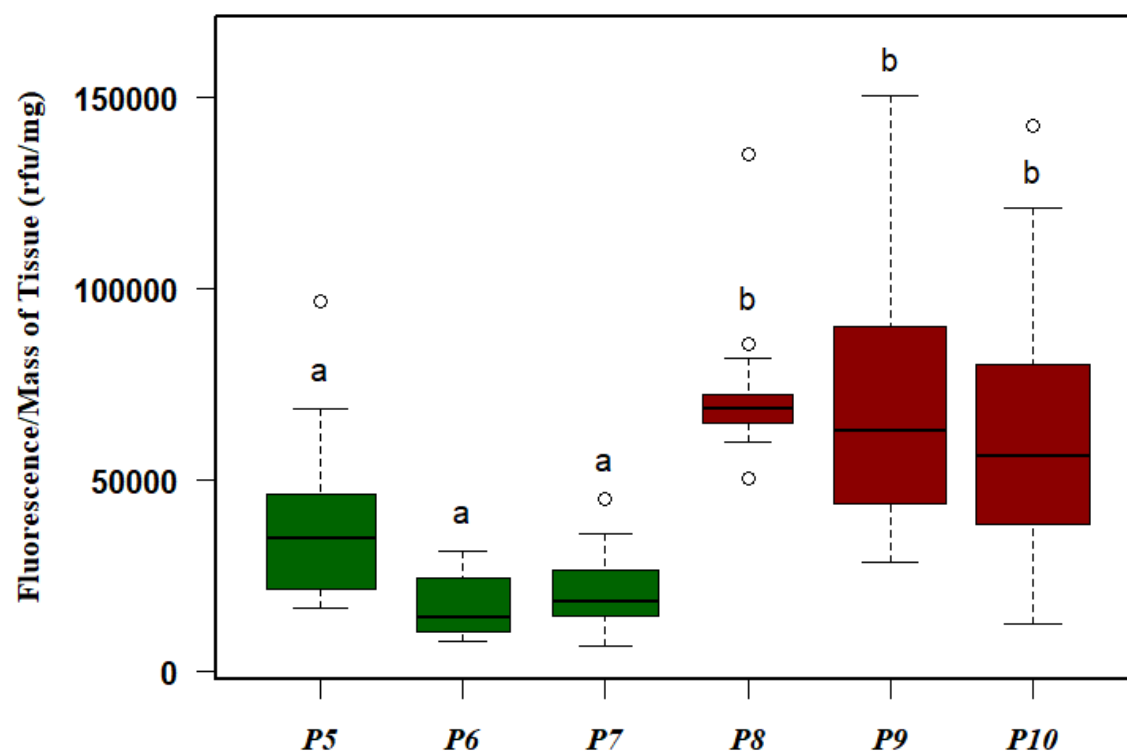


Figure 9: Boxplot with the MXR phenotype activity data of fish collected from sites inside (green) and outside the REBIO Guaribas (red), with its respective grouping by similarity above each box.

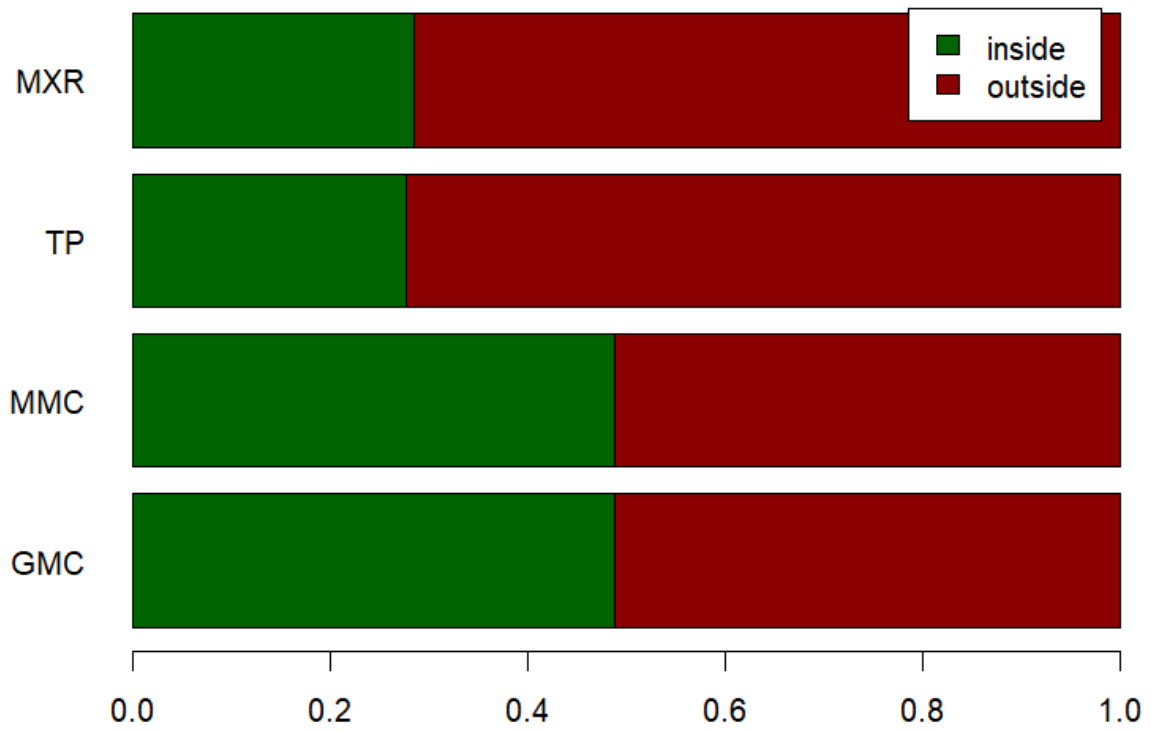


Figure 10: Proportion between average values for each physiological marker analysed, comparing themselves in order to its relationship with the REBIO Guaribas, highlighting the disparity among data of total protein (“TP”) and the MXR phenotype activity (“MXR”), and higher proximity among data of muscle (“MMC”) and gill moisture content (“GMC”).

Table 1: Table abundance.

```

1 library(dplyr)
2 library(tidyr)
3 library(gt)
4 library(openxlsx)
5
6 mxr23 <- read.xlsx("C:/Users/ellen/Downloads/Manuscritos/mxr23_Q/data/rebio23-peixes_5-10.xlsx",
7                   rowNames = T,
8                   colNames = T,
9                   sheet = "peixes-fisio")
10
11 data <- mxr23
12 data <- data[!row.names(data) %in% c("4-9-85", "1-10-7"), ]
13
14 data[c("1-10-4", "1-10-5"), c("Rodamina_rfu_mg", "TH_musculo_p",
15                             "TH_branquia_p", "Proteinas_Totais_mg")] <- NA
16
17 resumo2 <- data %>%
18   group_by(Site = Site, Espécie = Espécie, Amostra_tipo = Amostra_tipo) %>%
19   summarise(Count = n(), .groups = "drop") %>%
20   arrange(Site, Espécie, Amostra_tipo)
21
22 resumo2_wide <- resumo2 %>%
23   pivot_wider(
24     names_from = c(Site, Amostra_tipo),
25     values_from = Count,
26     values_fill = list(Count = 0)
27   )
28
29 resumo2_wide %>%
30   gt() %>%
31   cols_label(
32     Espécie = "Species"
33   ) %>%
34   fmt_number(
35     columns = everything(),
36     decimals = 0
37   ) %>%
38   tab_style(
39     style = cell_text(style = "italic"),
40     locations = cells_body(columns = Espécie)
41   ) %>%
42   tab_style(
43     style = cell_text(weight = "bold"),
44     locations = cells_column_labels()
45   ) %>%
46   opt_horizontal_padding(scale = 2) %>%
47   opt_table_lines("default")

```

Apendices

Non-used figures and tables

 Non-used codes